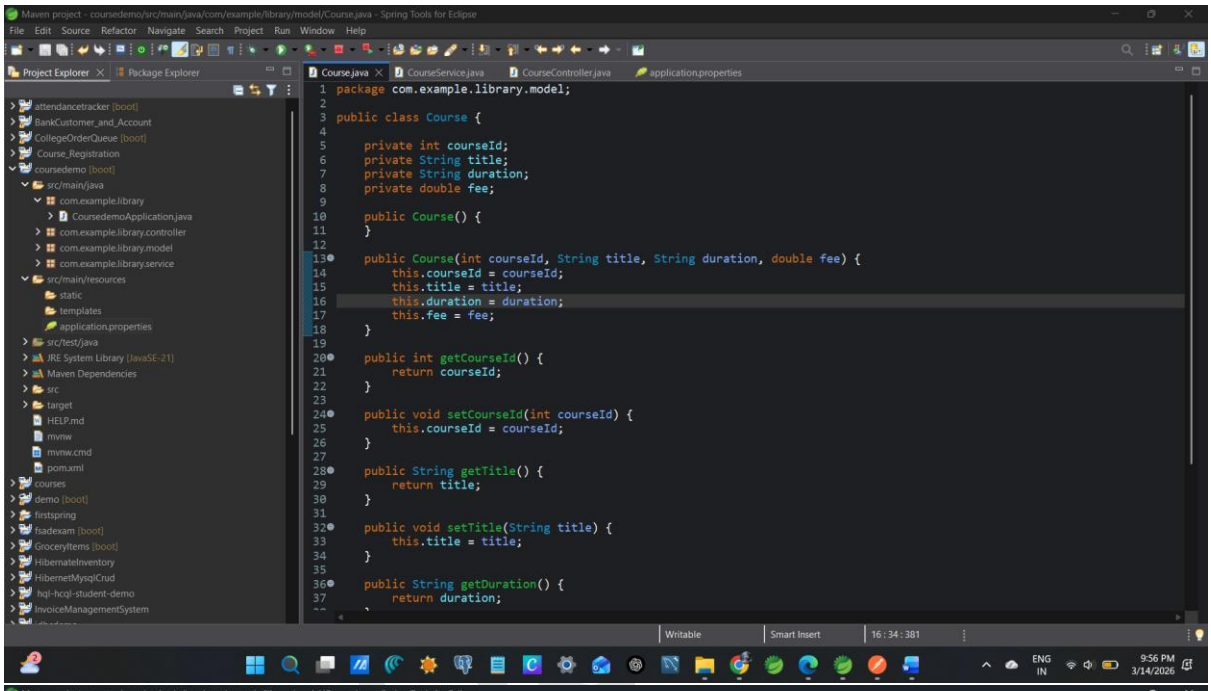
SKILL EXPERIMENT-7

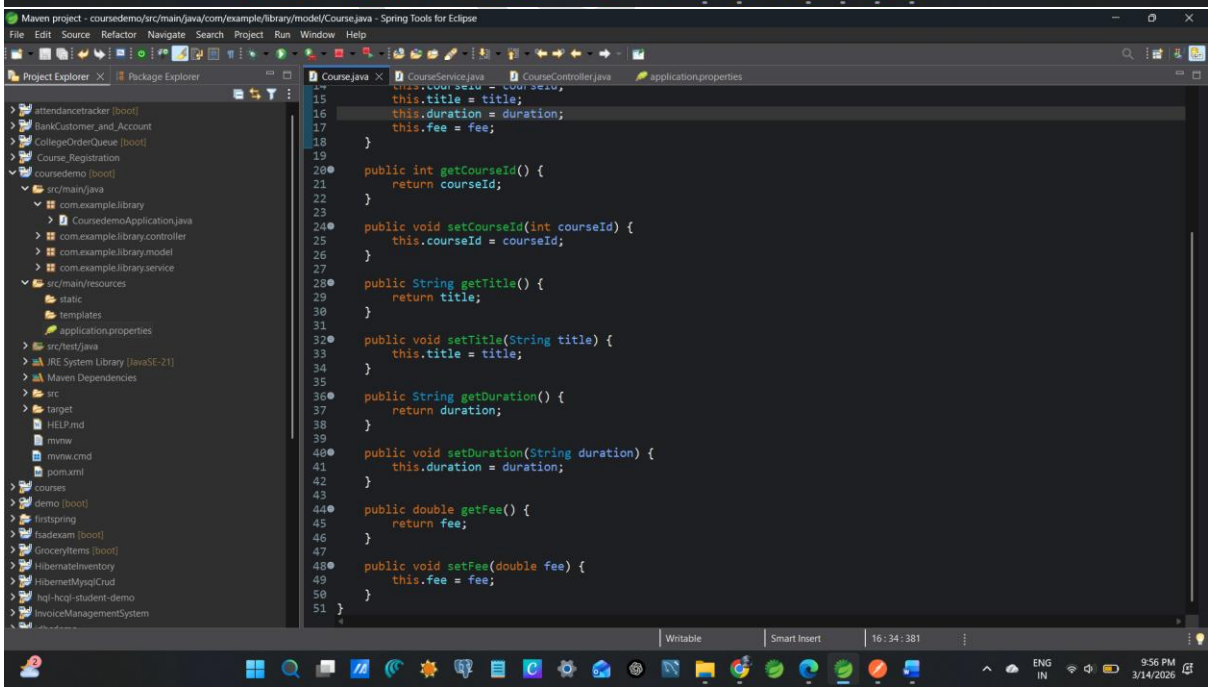
REST API - CRUD Operations using ResponseEntity

T. Amrutha Hasini

2400030474

FSAD, Sec 4





Maven project - coursedemo/src/main/java/com/example/library/service/CourseService.java - Spring Tools for Eclipse

```java
package com.example.library.service;

import com.example.library.model.Course;
import org.springframework.stereotype.Service;
import java.util.ArrayList;
import java.util.List;

@Service
public class CourseService {

    List<Course> courses = new ArrayList<>();

    public List<Course> getAllCourses() {
        return courses;
    }

    public Course addCourse(Course course) {
        courses.add(course);
        return course;
    }

    public Course updateCourse(int id, Course newCourse) {

        for (Course c : courses) {
            if (c.getCourseId() == id) {

                c.setTitle(newCourse.getTitle());
                c.setDuration(newCourse.getDuration());
                c.setFee(newCourse.getFee());

                return c;
            }
        }

        return null;
    }
```

Maven project - coursedemo/src/main/java/com/example/library/service/CourseService.java - Spring Tools for Eclipse

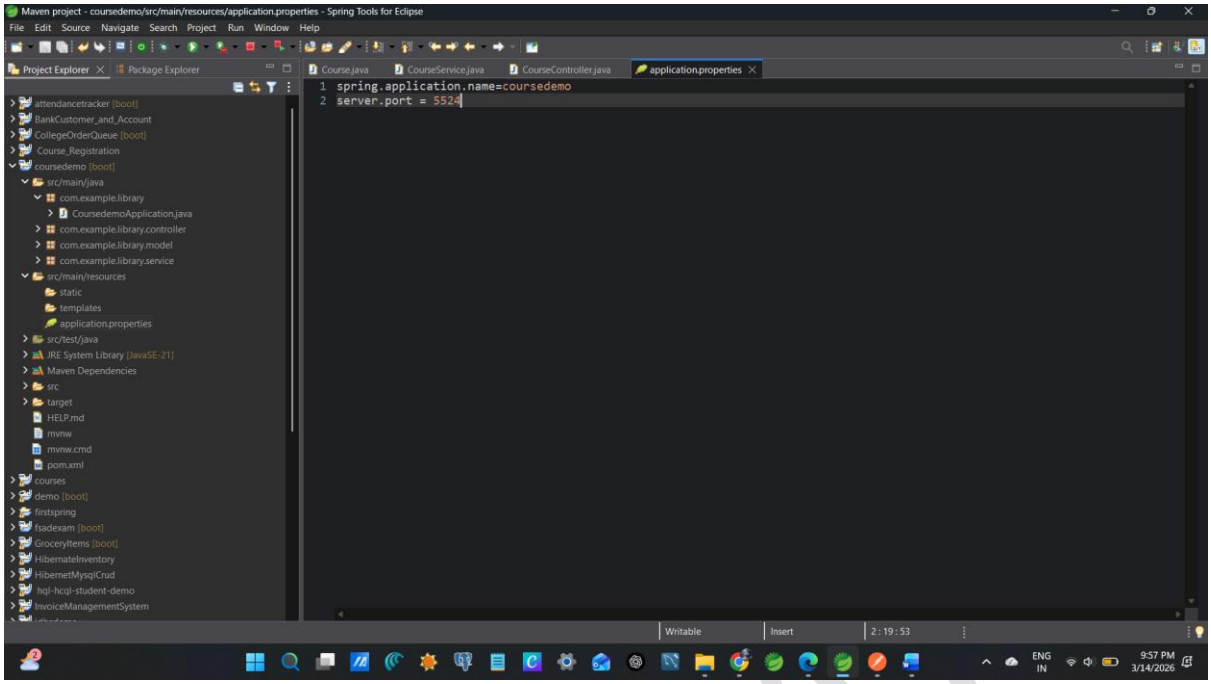
```java
        courses.add(course);
        return course;
    }

    public Course updateCourse(int id, Course newCourse) {

        for (Course c : courses) {
            if (c.getCourseId() == id) {

                c.setTitle(newCourse.getTitle());
                c.setDuration(newCourse.getDuration());
                c.setFee(newCourse.getFee());

                return c;
            }
        }

        return null;
    }

    public boolean deleteCourse(int id) {
        return courses.removeIf(c -> c.getCourseId() == id);
    }

    public List<Course> searchByTitle(String title) {

        List<Course> result = new ArrayList<>();

        for (Course c : courses) {
            if (c.getTitle().equalsIgnoreCase(title)) {
                result.add(c);
            }
        }

        return result;
    }
}
```

Maven project - coursedemo/src/main/java/com/example/library/controller/CourseController.java - Spring Tools for Eclipse

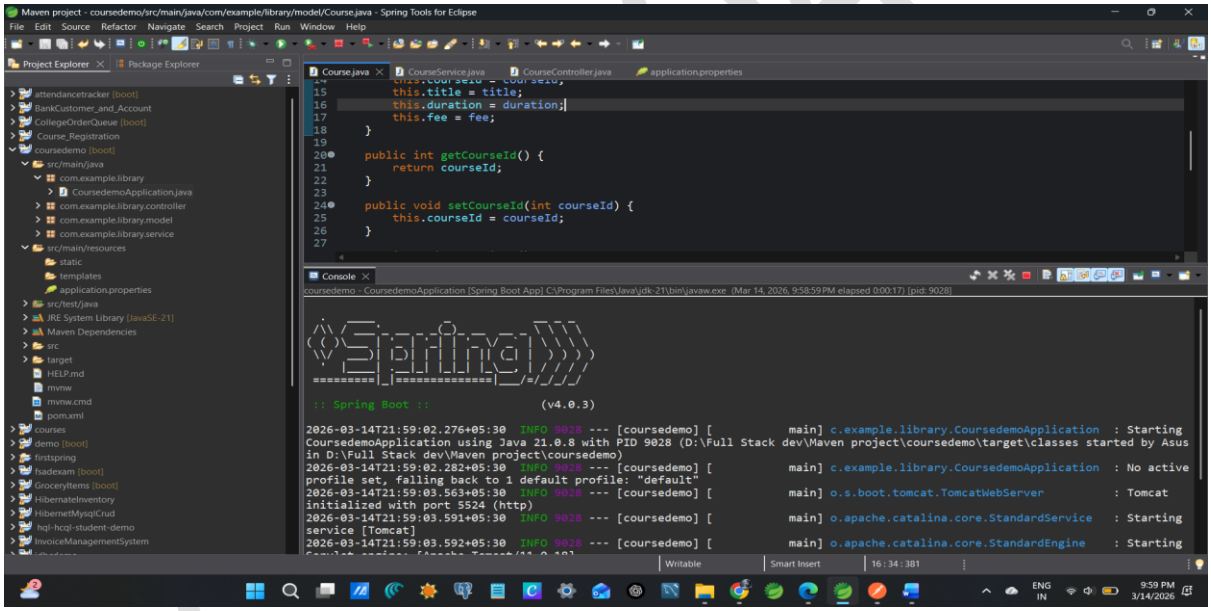
```java
package com.example.library.controller;

import com.example.library.model.Course;
import com.example.library.service.CourseService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.*;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
@RequestMapping("/courses")
public class CourseController {

    @Autowired
    CourseService service;

    @PostMapping
    public ResponseEntity<String> addCourse(@RequestBody Course course) {

        service.addCourse(course);
        return new ResponseEntity<>("Course Added Successfully", HttpStatus.CREATED);
    }

    @GetMapping
    public ResponseEntity<List<Course>> getCourses() {

        return new ResponseEntity<>(service.getAllCourses(), HttpStatus.OK);
    }

    @PutMapping("/{id}")
    public ResponseEntity<String> updateCourse(@PathVariable int id, @RequestBody Course course) {

        Course updated = service.updateCourse(id, course);

        if (updated == null) {
            return new ResponseEntity<>("Course Not Found", HttpStatus.NOT_FOUND);
```

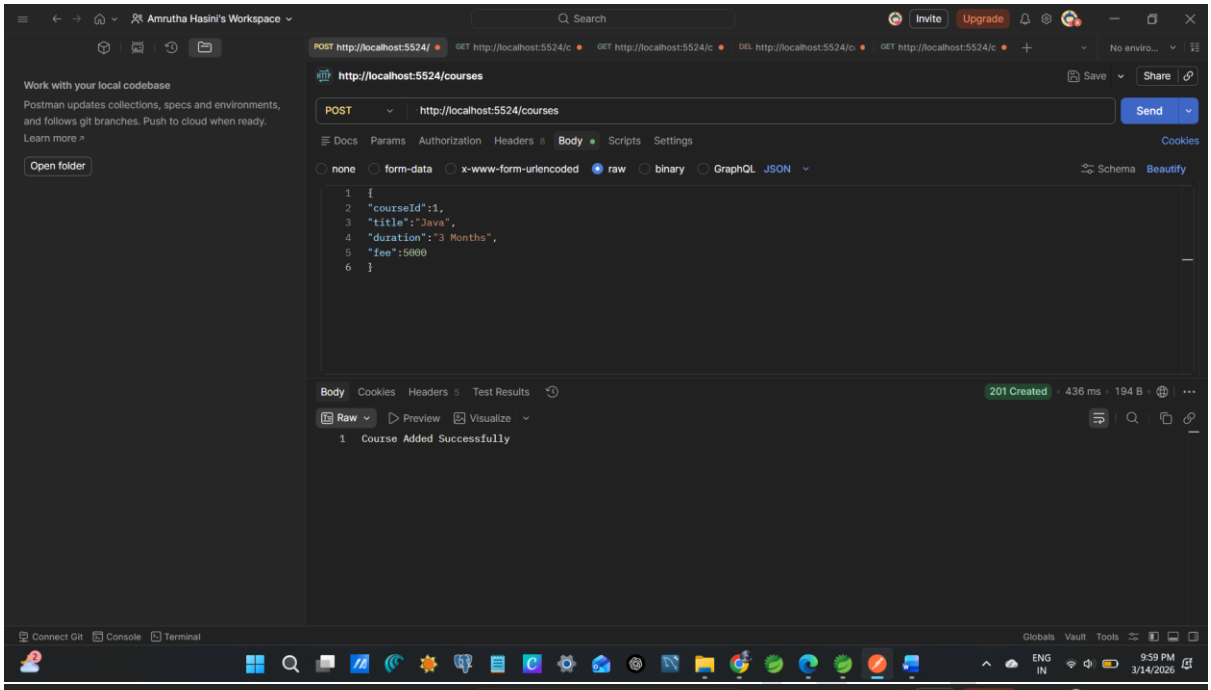
Maven project - coursedemo/src/main/java/com/example/library/controller/CourseController.java - Spring Tools for Eclipse

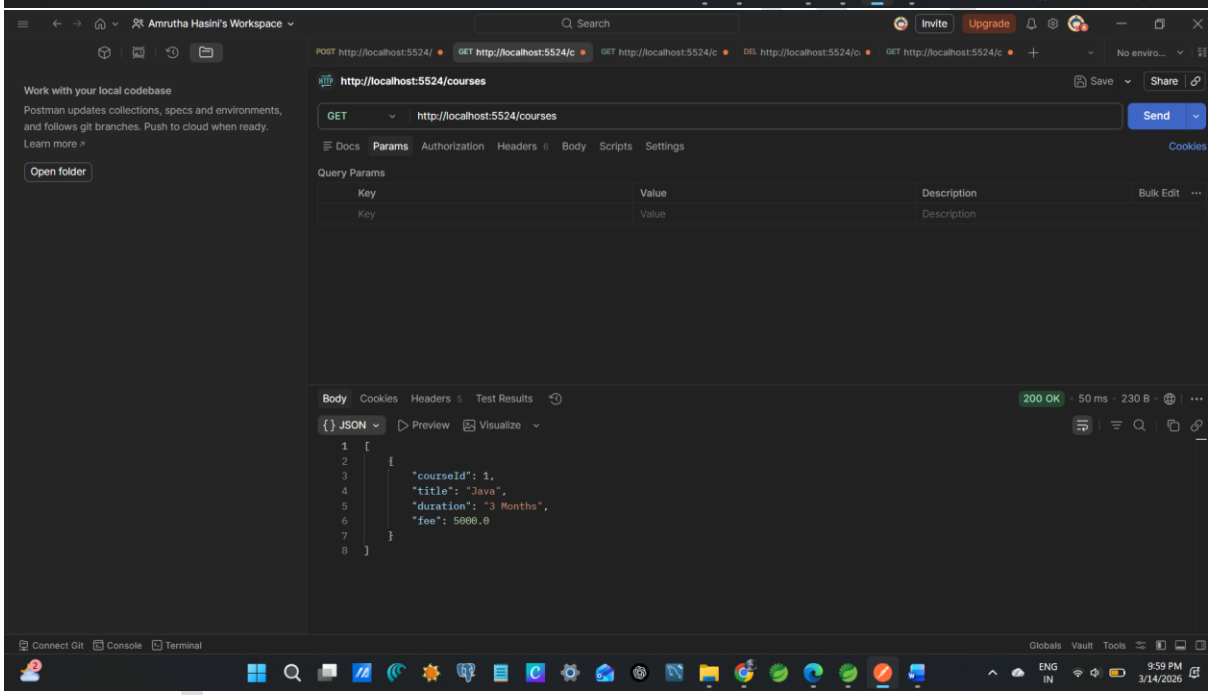
```java
    }

    @PutMapping("/{id}")
    public ResponseEntity<String> updateCourse(@PathVariable int id, @RequestBody Course course) {

        Course updated = service.updateCourse(id, course);

        if (updated == null) {
            return new ResponseEntity<>("Course Not Found", HttpStatus.NOT_FOUND);
        }

        return new ResponseEntity<>("Course Updated Successfully", HttpStatus.OK);
    }

    @DeleteMapping("/{id}")
    public ResponseEntity<String> deleteCourse(@PathVariable int id) {

        boolean deleted = service.deleteCourse(id);

        if (!deleted) {
            return new ResponseEntity<>("Course Not Found", HttpStatus.NOT_FOUND);
        }

        return new ResponseEntity<>("Course Deleted Successfully", HttpStatus.OK);
    }

    @GetMapping("/search/{title}")
    public ResponseEntity<?> searchCourse(@PathVariable String title) {

        List<Course> list = service.searchByTitle(title);

        if (list.isEmpty()) {
            return new ResponseEntity<>("No Course Found", HttpStatus.NOT_FOUND);
        }

        return new ResponseEntity<>(list, HttpStatus.OK);
    }
}
```

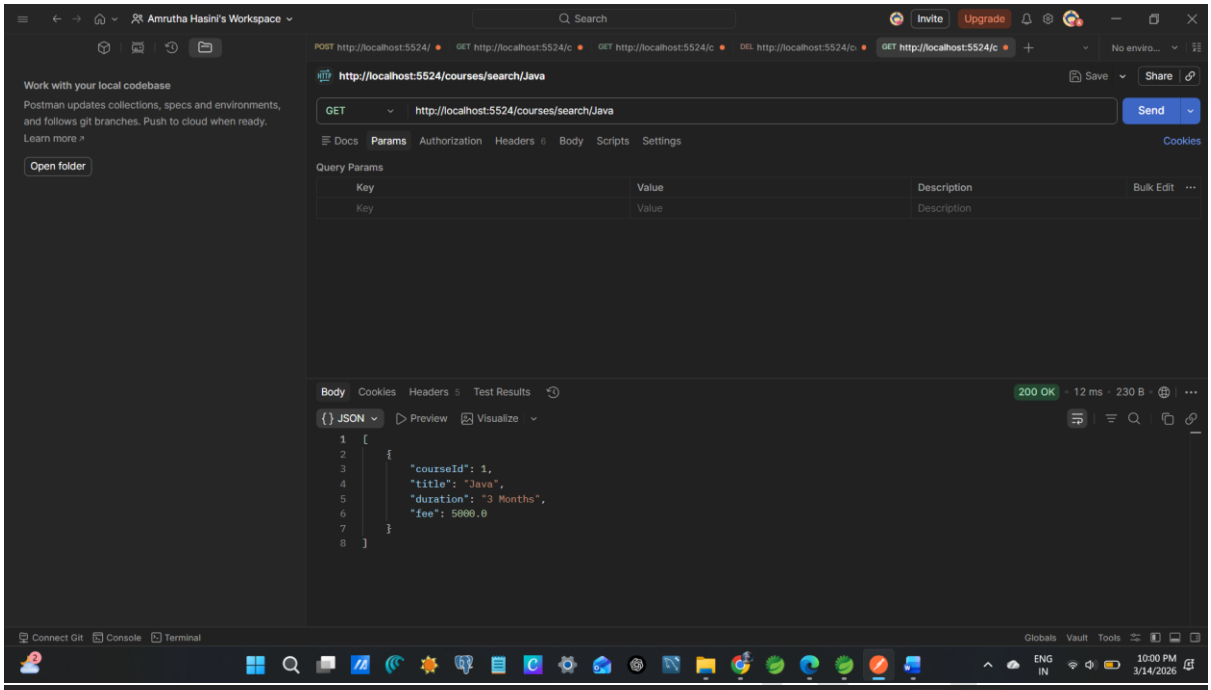

Maven project - coursedemo/src/main/resources/application.properties - Spring Tools for Eclipse
File Edit Source Navigate Search Project Run Window Help
Project Explorer
Package Explorer
Course.java
CourseService.java
CourseController.java
application.properties
spring.application.name=coursedemo
server.port = 5524
attendancetracker [boot]
BankCustomer_and_Account
CollegeOrderQueue [boot]
Course_Registration
coursedemo [boot]
src/main/java
com.example.library
CoursedemoApplication.java
com.example.library.controller
com.example.library.model
com.example.library.service
src/main/resources
static
templates
application.properties
src/test/java
JRE System Library [JavaSE-21]
Maven Dependencies
src
target
HELP.md
mvnw
mvnw.cmd
pom.xml
courses
demo [boot]
firstspring
fsadexam [boot]
GroceryItems [boot]
HibernateInventory
HibernetMysqlCrud
hql-hcql-student-demo
InvoiceManagementSystem
Writable
Insert
2 : 19 : 53
ENG
IN
9:57 PM
3/14/2026

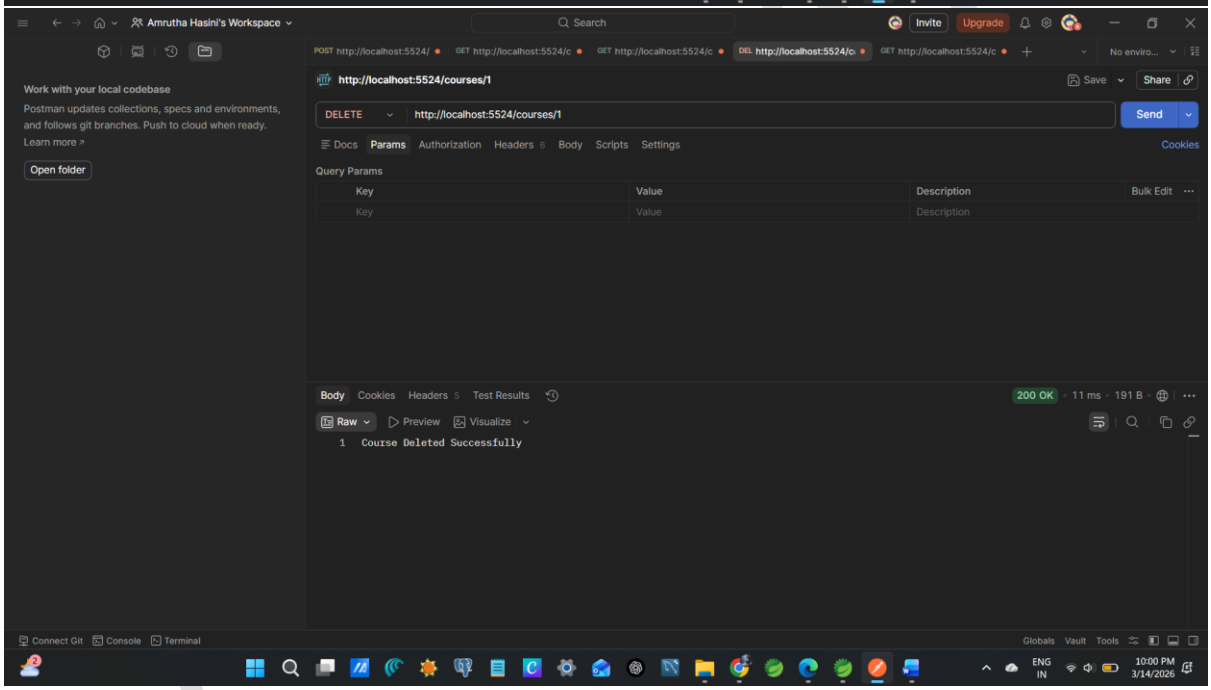
OUTPUT: -


Maven project - coursedemo/src/main/java/com/example/library/model/Course.java - Spring Tools for Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
Package Explorer
Course.java
CourseService.java
CourseController.java
application.properties
this.title = title;
this.duration = duration;
this.fee = fee;
}
public int getCourseId() {
return courseId;
}
public void setCourseId(int courseId) {
this.courseId = courseId;
}
Console
coursedemo - CoursedemoApplication [Spring Boot App] C:\Program Files\Java\jdk-21\bin\javaw.exe (Mar 14, 2026, 9:58:59 PM elapsed 0:00:17) [pid: 9028]
:: Spring Boot :: (v4.0.3)
2026-03-14T21:59:02.276+05:30 INFO 9028 --- [coursedemo] [main] c.example.library.CoursedemoApplication : Starting CoursedemoApplication using Java 21.0.8 with PID 9028 (D:\Full Stack dev\Maven project\coursedemo\target\classes started by Asus in D:\Full Stack dev\Maven project\coursedemo)
2026-03-14T21:59:02.282+05:30 INFO 9028 --- [coursedemo] [main] c.example.library.CoursedemoApplication : No active profile set, falling back to 1 default profile: "default"
2026-03-14T21:59:03.563+05:30 INFO 9028 --- [coursedemo] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 5524 (http)
2026-03-14T21:59:03.591+05:30 INFO 9028 --- [coursedemo] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-03-14T21:59:03.592+05:30 INFO 9028 --- [coursedemo] [main] o.apache.catalina.core.StandardEngine : Starting
Writable
Smart Insert
16 : 34 : 381
ENG
IN
9:59 PM
3/14/2026


Amrutha Hasini's Workspace
Search
Invite
Upgrade
Work with your local codebase
Postman updates collections, specs and environments, and follows git branches. Push to cloud when ready.
Learn more
Open folder
http://localhost:5524/courses
Save
Share
POST
http://localhost:5524/courses
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"courseId":1,
"title":"Java",
"duration":"3 Months",
"fee":5000
}
Body
Cookies
Headers 5
Test Results
201 Created
436 ms
194 B
Raw
Preview
Visualize
Course Added Successfully
Connect Git
Console
Terminal
Globals
Vault
Tools


Amrutha Hasini's Workspace
Search
Invite
Upgrade
Work with your local codebase
Postman updates collections, specs and environments, and follows git branches. Push to cloud when ready.
Learn more
Open folder
http://localhost:5524/courses
Save
Share
GET
http://localhost:5524/courses
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 5
Test Results
200 OK
50 ms
230 B
JSON
Preview
Visualize
[
{
"courseId": 1,
"title": "Java",
"duration": "3 Months",
"fee": 5000.0
}
]
Connect Git
Console
Terminal
Globals
Vault
Tools





GitHub repo:- https://github.com/AmruthaHasini-KLU/skill-7.git